Alex Towell
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Alex Towell

PhD Student | AI Researcher | Mathematician

About Me I am a PhD student in Computer Science at Southern Illinois University, focusing on artificial intelligence, AI safety, and alignment. My work spans over 100 open-source projects exploring novel approaches in mathematics, cryptography, and AI. As an autodidact, I've independently studied advanced topics in physics, philosophy, and literature. I seek to join a leading AI research lab to collaborate on challenging problems that have real implications for the future.

Education

2024 - Present, Southern Illinois University

PhD in Computer Science

- Research Focus: Cybersecurity, AI Safety, Alignment, Machine Learning
- Expected Graduation: 2028

2021 - 2023, Southern Illinois University-Edwardsville

MS in Mathematics

- GPA: 3.9/4.0
- Concentration: Statistics and Operations Research
- Awarded R.N. Pendergrass Graduate Scholarship

2012 - 2015, Southern Illinois University-Edwardsville

MS in Computer Science

- GPA: 4.0/4.0
- Concentration: Artificial Intelligence
- Awarded Outstanding Graduate Student (School of Engineering)

2006 - 2010, Southern Illinois University-Edwardsville

BS in Computer Science

- GPA: 3.6/4.0
- Awarded Outstanding Junior (Dept. of Computer Science)

Technical Skills

Languages

Python, R, C/C++, Java, Scala, JavaScript, Bash

AI/ML

PyTorch, HuggingFace, CUDA, LangChain, Fine-Tuning

Tools

Linux, Git, Docker, VS Code, Jupyter, LaTeX, Markdown

Cryptography

Encryped Search, Secure Indexes, Oblivious Computing

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Publications

- "Reliability Estimation in Series Systems: Maximum Likelihood Techniques for Right-Censored and Masked Failure Data," Southern Illinois University-Edwardsville Department of Mathematics and Statistics, Masters Project, Oct 2023
- "Estimating how confidential encrypted searches are using moving average bootstrap method," IEEE Cloudcom, Dec 2016
- "Encrypted search: enabling standard information retrieval techniques using new secure index types while preserving confidentiality against an adversary with access to hidden query histories and contents of secure indexes," ProQuest Dissertations & Theses, Aug 2015

Selected Research & Projects

AlgoTree

Developed a Python package offering utilities for tree-like data structures, including FlatForest and TreeNode representations. Features a command-line tool `jt` for creating, manipulating, querying, and visualizing trees, akin to `jq` but tailored for hierarchical data.

Elasticsearch-LM

Initiated a project to fine-tune smaller Large Language Models (LLMs) for translating natural language queries into structured Elasticsearch Query DSL. Emphasizes synthetic data generation to enhance model training, aiming to democratize sophisticated API interactions.

Reverse-Process Synthetic Data Generation

Developed algorithms generating synthetic data via reverse-process methods to facilitate LLM-based mathematical reasoning and theorem proving.

Oblivious Computation

Researched cryptographic methods enabling secure computation trade-offs between space and security without full homomorphism.

Academic Experience

2025–Present, Research Assistant, SIU

- Research in cybersecurity under Dr. Fujinoki, with a focus on ransomeware detection and prevention.
- Full-stack development for IDOT-D8.

2021–2022, Teaching Assistant, SIU

- Taught lab courses in mathematics and statistics.